RIYA SANDESH SHINDE

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PROFESSIONAL SUMMARY

Ph.D. candidate in Chemical & Biochemical Engineering specializing in **Pharmaceutical process development, Continuous Manufacturing of Solid Dose, and Formulation Development**. Extensive understanding of **Amorphous Solid dispersion** with experience in **Hot melt extrusion, Melt and Wet Granulation**. Demonstrated track record of leading cross-functional research projects, mentoring junior researchers, and managing laboratory operations. Deep expertise in **Design of Experiments (DoE), Quality by Design (QbD)**, and advanced analytics applied to oral solid dose manufacturing. Recognized speaker at AIChE and active contributor to industry consortia. cGMP experience.

WORK EXPERIENCE

Ph.D. Researcher | C-SOPS, Rutgers University – New Jersey *Sep 2022 – Present*

Advisor: Dr. Fernando Muzzio | Focus: **Formulation R&D of Oral Solid Dose Pharmaceuticals**, process development, and PAT

- Improved feedability of highly cohesive and adhesive APIs through dry coating in collaboration with ***Vertex Pharmaceuticals***— presented at AIChE Annual Meeting 2023
- Enhanced processability and solubility of poorly soluble drugs via Advanced Manufacturing Technology (AMT) - **melt coating granulation** for both batch and continuous processes in collaboration with ***Johnson and Johnson Innovative Medicine***
- Developed a novel rapid tech transfer approach from benchtop to continuous manufacturing using platform formulations and PCA-based surrogate matching in collaboration with ***Johnson and Johnson Innovative Medicine***
- Developed shear-independent platform formulations for rapid transfer from benchtop to continuous manufacturing in collaboration with ***Merck KGaA***
- Utilized PCA to identify property-matched material surrogates for pharmaceutical and catalyst applications
- Executed **DoE** and **QbD** frameworks for oral solid dose manufacturing; applied NIR and Raman spectroscopy for non-destructive blend characterization
- Oversaw laboratory operations, safety compliance, equipment and procurement, secure data management, industry–academic collaborations, and graduate student training in experimental design and data integrity

Formulations & Development Co-Op | Bayer Consumer Health – New Jersey *Jan 2026 – Jun 2026*

- Explored capabilities of the **Leistritz ZSE 12 twin-screw extruder** for pharmaceutical process development
- **Stabilized moisture-sensitive APIs** using melt coating granulation (MCG) to enhance stability, formulation performance, and taste masking
- Expanded work to a novel drug form for compressible gummies (Soft Solids) via Wet extrusion for continuous manufacturing

Research Assistant | C-SOPS, Rutgers University — New Jersey *Feb 2022 – Aug 2022*

- Characterized raw and processed powders using Freeman Technology FT4 Rheometer and laser diffraction particle size analysis (LD-PSA)

- Presented powder flow and handling data at the bi-annual Catalyst Manufacturing Consortium

Graduate Engineer Trainee | SAGA Chemie Pvt. Ltd — Maharashtra, India *Jun 2019 – Jan 2021*

- Supported production planning, materials procurement, and quality control improvements in compliance with safety and regulatory standards

EDUCATION

Ph.D., Chemical & Biochemical Engineering (Pharmaceutical Sciences)
Rutgers, The State University of New Jersey · GPA: 3.55 *Expected 2026*
Technology Transfer Traineeship (Aug 2025)

M.S., Chemical & Biochemical Engineering
Rutgers, The State University of New Jersey · GPA: 3.55 *Aug 2022*

B.S., Chemical Engineering
University of Mumbai, India · GPA: 3.5 *Jul 2019*

TECHNICAL SKILLS & EQUIPMENT

Process Equipment	Leistritz ZSE 12 Twin-Screw Extruder, Thermo Pharma 11 & 16 (MCG/Wet Granulation), Coperion K-Tron Feeder, Fette 1200 & Roltgen FlexiTab Tablet Presses, Adamus One Station, LabRAM/V-Blender/Co-axial/Bin Blenders, High Shear Mixer
Characterization	FT4 & Brookfield Rheometers, Anton Paar Rheometer, LD-PSA, Texture Analyzer, Differential Scanning Calorimetry, Thermogravimetric Analysis, Hardness Testing, Dissolution & Drug Release (UV-Vis), UHPLC/HPLC, NIR (MPA Bruker), Raman Spectroscopy, GAMLEN Tablet Press, STYL'One Compaction simulator
Software & Data	JMP, SIMCA, MATLAB, Origin, Microsoft Office, Adobe Illustrator, AutoCAD, Empower
Soft Skills	Cross-functional project leadership, lab management, graduate mentorship, conference presentation, industry consortium collaboration

SELECTED PUBLICATIONS (IN PREPARATION)

- Shinde R.S., Vyas Z.D., Scicolone J., Muzzio F.J. — "Shear sensitivity of melt coated granules on compressibility and dissolution characteristics"
- Shinde R.S., Scicolone J., Muzzio F.J. — "Systematic study on improving flowability of poorly flowing API by dry coating"
- Vyas Z.D., Shinde R.S., Scicolone J., Muzzio F.J. — "Twin-Screw Melt Coating and Granulation to enhance bioavailability of Fenofibrate"
- Shinde R.S., Scicolone J., Callegary G., Muzzio F.J. — "Exploring effects of shear for rapid transfer from benchtop to continuous manufacturing"
- Shinde R.S., Scicolone J., Muzzio F.J. — "Using platform formulations for rapid transfer from benchtop to continuous manufacturing"